NIRMAL KUMAR THANGARAJU

SENIOR SDET | QUALITY ENGINEER

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Professional Summary

Senior SDET / Quality Engineer with 8 years of experience designing and scaling robust UI and API test automation for enterprise healthcare and government platforms. Strong expertise in TypeScript, Playwright, Postman, GitHub Actions, and Docker, delivering a 70% reduction in manual regression effort, increasing automated test coverage. Adept at integrating automated testing into CI/CD pipelines to strengthen release confidence and accelerate delivery. Experienced in healthcare quality-reporting systems, HIPAA-aware testing, and Section 508/WCAG accessibility validation, with a versatile approach to quality engineering across regulated and non-regulated industries.

Technical Skills

Languages & Version Control: TypeScript, JavaScript, SQL, Git
E2E Automation: Playwright, Cypress, Page Object Model, Custom Fixtures, Parallel Execution, Cross-browser Testing, Trace Viewer
API Testing: Postman, REST APIs, GraphQL, Swagger UI, Authentication, Schema Validation, API Chaining
Test Data Management: JSON, API-Driven Test Data Setup, Reusable Test Fixtures, Test Data Cleanup, API Collections
Accessibility Testing: WAVE, NVDA, axe-core
Performance Testing: K6, Browser DevTools
CI/CD & DevOps: GitHub Actions, Docker, CI/CD Pipeline Development, Automated Test Reporting
Test Management: Jira, Confluence, Zephyr, TestRail, Requirement Traceability Matrix (RTM)
Methodologies: Agile (Scrum), Scaled Agile Framework (SAFe), Lean Six Sigma

Professional Experience

SemanticBits LLC **USA & India**

Senior QA Engineer | *Projects: Medicaid and CHIP (MAC) & Measure Authoring Tool (MAT)* **Mar 2020 – Feb 2025**

- Led end-to-end QA strategy for Medicaid and CHIP (MAC), mentoring junior QAs and UAT testers while achieving federal HIPAA compliance and accelerating UAT sign-off by 25%.
- Architected and scaled a Playwright + TypeScript automation framework using Page Object Model, reusable custom fixtures, and API-driven test workflows, increasing automated test coverage by 60% through cross-browser testing.
- Utilized Postman to validate the APIs feeding D3.js visualizations, verifying count accuracy and catching data mismatches.
- Engineered containerized Playwright test execution using custom Docker images, GitHub Actions CI/CD, and parallel regression execution, reducing CI-only environment failures by 75% and accelerating defect triage through automated reports and traces.
- Developed Cypress + JavaScript E2E automation for the Measure Authoring Tool (MAT), automating CQL Expression Editor, measure packaging, and export workflows while reducing manual regression effort by 70%.
- Validated eCQM workflows across HQMF, CQL, and ELM formats, collaborating with CMS stakeholders, measure developers, and analytics teams to identify 25% more defects before release through RTM-driven coverage analysis.
- Performed WCAG accessibility validation using axe-core, NVDA and WAVE, testing keyboard navigation, screen-reader behavior, and accessibility compliance across enterprise workflows.

IT Automation LLC **USA**

Quality Assurance Engineer | *Project: Quality Payment Program (QPP)* **Feb 2017 – Feb 2020**

- Designed and executed smoke, sanity, regression, functional, integration, and non-functional testing to ensure product quality.
- Architected Cypress E2E automation from scratch, migrating from Protractor/Selenium and improving testing efficiency by 40%.
- Automated MIPS data submission and validation workflows across JSON, XML, and QRDA III formats, including feedback-report downloads, reducing processing errors by 60%.
- Developed Postman API automation collections for complex scoring endpoints, schema validations, authentication, API chaining, and user-profile creation workflows backed by MongoDB, increasing API test coverage by 35%.
- Led Section 508 and WCAG 2.0 accessibility validation using JAWS screen reader and BrowserStack for cross-platform testing.
- Managed test strategy, test cases, execution, defects, and QA documentation using Jira, Zephyr, and Confluence.

Planned Career Sabbatical | *Family Caregiving* **Mar 2025 – Present**

- Took a planned sabbatical for family caregiving while focusing on professional development in AI-assisted quality engineering and test automation.

Education and Certification

Master of Science in Technology – *University of Central Missouri* **Aug 2015 – Dec 2016**

Graduate Certificate in Lean Six Sigma – *American Society for Quality (ASQ)* **Issued: Dec 2016**